MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

Discipline Specific Elective Course (DSE)

MG4DSEBOTA03: ELEMENTARY PRINCIPLES AND APPLICATIONS OF PLANT BIOTECHNOLOGY & INDUSTRIAL BOTANY

(2024 ADMISSION ONWARDS)

Time: 2 Hrs **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

One or two Sentences

Answer any ten questions Each question carries 2 marks

1.	What are cryoprotectants?	[K]	[2]
2.	Name any two disadvantages of transgenic plants.	[U]	[2]
3.	What is rDNA technology?	[K]	[2]
4.	Define proteomics.	[K]	[1]
5.	How would you summarize the types of molecular markers?	[U]	[4]
6.	Recall the use of biotechnology in environmental monitoring.	[K]	[3]
7.	Name any four plants used in biofuel production	[K]	[6]
8.	Why are biopolymers modified?	[A]	[6]
9.	What are gene banks?	[U]	[5]
10.	Enlist any two plants used in bioremediation.	[K]	[5]
11.	Short note on biostimulation.	[U]	[5]
12.	Comment on the importance of ginger house.	[U]	[5]

(2 × 10 = 20)

PART - B

Short answer

Answer any eight questions. Each question carries 5 marks

13. Elucidate the applications of somatic cell hybridization in plant improvement.	[K]	[1]
14. Describe the steps involved in cryopreservation.	[K]	[2]
15. Discuss the importance of transgenics in plant biotechnology.	[U]	[2]
16. What are the applications of biotechnology in waste management.	[A]	[3]
17. Summarize the significance of marker-assisted breeding in crop improvement citing an example.	[An]	[4]
18. What are the applications of natural dyes extracted from plant sources?	[A]	[6]
19. Explain with examples the different types of bioplastics.	[K]	[6]
20. How can resource management be optimized in bioremediation?	[A]	[5]
21. How can industrial botany be applied in the restoration of degraded habitats? Discuss with suitable examples.	[A]	[5]
22. Comment in detail the importance of biodiversity conservation for industrial applications	[An]	[5]

(5 × 8 = 40)

PART – C

Essay

Answer any one question. Each question carries 10 marks

23. What are essential oils? Explain in detail the process of extracting essential oils from plant sources.	[U]	[6]
24. Write an essay on the application of biotechnology in improving agriculture. Also discuss the ethical issues raised in this regard.	[A]	[3]

(10 × 1 = 10)